EMBEDDED / CONTROL / AUTONOMY

AEGIS AUTONOMY

02

PROPOSED RESEARCH / NOT YET VERIFIED

THE ENGINEERING QUESTION

Which monitoring and degraded control strategies reduce risk without triggering unusable false alarms when state estimates become unreliable?

PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A fault aware autonomy research platform investigating whether a vehicle can recognise deteriorating state trust and degrade safely before control decisions become unsafe.

- This is not another generic drone simulator; the differentiator is uncertainty aware supervision.
- The value comes from controlled experiments, negative results and honest limits, not pretty trajectories.
- It extends your existing drone background into research grade verification.

ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE **Vehicle dynamics** → 2 STAGE **Estimator** → 3 STAGE **Fault monitor** → 4 STAGE **Degraded controller**

- C++ estimator and control core with reproducible configuration sets and seeded Monte Carlo runs.
- Error state or EKF estimation, covariance monitoring, residual analysis, supervisory states and bounded degraded operation.
- Separate assumptions, dynamics model, fault generation logic and evaluation harness so the work is inspectable.

ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ Simulation robustness is not equivalent to hardware safety certification.

⚠ Detectability depends on observability and the validity of the assumed sensor/vehicle models.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

AEGIS AUTONOMY / PROTOCOL / ACCEPTANCE / REVIEW

EXPERIMENTS, VERIFICATION & PUBLICATION

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EXPERIMENTAL METHODOLOGY

01 Build deterministic 6 DoF or reduced order dynamics with IMU/GPS noise, bias, delay and dropout.

02 Compare fixed threshold, innovation based and adaptive monitoring strategies against a plain baseline controller.

03 Measure detection delay, false alarms, trajectory error, constraint violations and mission completion under varied faults.

MEASUREMENT AND ACCEPTANCE EVIDENCE

- **Fault detection delay and false alarm rate**
- **Trajectory error and safety boundary violations**
- **Sensitivity to estimator miscalibration and correlated faults**

REQUIRED REPOSITORY ARTEFACTS

- **Reproducible experiment runner**
- **Baseline and monitored controller benchmark report**
- **Technical brief explaining where the method fails**

TECHNICAL REVIEW QUESTIONS

Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?

VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.